

Fazul Al Rehman

Offensive Security Researcher | Bug Bounty Hunter

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EDUCATION

BS Computer Science | Institute of Business Administration (IBA), Karachi

2021 – 2025

Final Year Project: IntelliScore, AI-Based Credit Scoring System (in collaboration with Meezan Bank)

HSSC (Pre-Engineering) | Superior College of Sciences

2019

PROFESSIONAL EXPERIENCE

Independent Security Researcher | Offensive Security & Bug Bounty

2019 – Present

Full-time bug bounty hunter since Jan 2026. Offensive security research and responsible disclosure across enterprise, e-commerce, gaming, and Web3 targets.

Samsung Pass, Critical Production Vulnerability

Critical, Bugcrowd

- Broke access control to reach internal environment secrets, including a **Jenkins CI/CD Slack** token with admin-level access to sensitive operational channels.
- Mapped live dashboards, **Terraform**, SSH key rotations and internal credentials, and gained further controls leading directly to **Samsung's internal databases and servers**; disclosed via **Bugcrowd** for a **\$10k bounty** on a biometric platform affecting **100M+ users**.

Walmart Marketplace, Payment Access-Control Bypass

Critical

- Critical broken-access-control flaw in **Walmart Marketplace Seller Central**: any authenticated account could reroute incoming funds on **any** merchant account, leaving all **100k+ Marketplace merchants** directly drainable, an estimated **\$500M+** in payment flow at risk.

Large-Scale Credential Exposure via Dorking

Disclosure in progress

- Systematic search-engine dorking surfaced **20,000+ exposed private keys** and API secrets across **AWS, Stripe, Twilio, Razorpay, OpenAI, Anthropic, Shopify, Twitch**, and **GitHub**.
- Aggregated **GitHub** keys exposed auto-deploying access to **1,600+ private repositories**; **Shopify** secret keys affected multiple e-commerce brands (**Shark Tank India**-featured startups, **ushopmall, Greyson Clothiers**). Coordinating responsible disclosure with affected organizations.

IBA Karachi, University Portal Security Assessment

Responsibly disclosed

- Identified multiple critical vulnerabilities across the portals of IBA Karachi, a top Pakistani university, and disclosed them to administration, prompting a full migration to a new portal now in production.
- Findings: a database exposure leaking confidential personal records (including stored .png files), and an authentication bypass reaching faculty accounts up to super-admin level.

EA Origin, Security-Question Authentication Bypass

Auth bypass

- Security-question bypass in **EA Origin's** account-recovery flow, enabling authentication without the legitimate answers.

Dell, Invoice-PDF Authorization Bypass

Auth bypass

- Authorization bypass on **Dell** allowing retrieval of invoice PDFs through serial-number enumeration.

Web3 Security Research

Ongoing

- Auditing smart contracts through **Cantina** and **Immunefi**.

Teaching Assistant | Software Architecture & Design

Aug 2025 – Dec 2025

- Mentored **50+ students** on software-architecture principles and design patterns, running tutorials that grounded the theory in real-world scenarios, including constraints like limited internet access and remote-area connectivity.
- Improved grades and conceptual understanding across the class; students consistently pointed to the tutorials as what made the architecture-design material click, a clear and lasting impact.

Mindsquared | CTO (Remote)

Jan 2024 – Jun 2024

- Built data pipelines driving profitability roadmaps across **Walmart** and **Amazon** operations; helped **50+ companies** reach profitability through automated analytics.

KEY PROJECTS

IntelliScore, AI Credit Scoring for **Meezan Bank**

Final Year Project, IBA Karachi

Led a four-person team from a one-line industry brief to a deployed, bank-facing AI product.

- Owned product direction for **Meezan Bank** (Pakistan's largest Islamic bank): scoped an undefined brief across six executive sessions, negotiated a data-sharing NDA, and absorbed three mid-flight scope pivots (consumer loans → MyBykea bike financing → auto financing) without slipping delivery, replacing a days-long manual ECIB review.
- Architected a **semi-supervised teacher-student distillation** pipeline over **93,000 anonymized records** and 32 features (research-backed formula labeled 50k, a fine-tuned **MLPRegressor** 128-64-32 pseudo-labeled 30k, a **HistGradientBoostingRegressor** trained across both), cutting validation MSE from 15.67 to **3.12** at $R^2 = 0.9971$ and holding $R^2 = 0.99$ held-out at **sub-2s** inference.
- Built a **RAG-grounded LLM** explainability chatbot and shipped the full platform, UI and backend both in **Python / Django**, over **PostgreSQL** and **Tailwind CSS**, with multi-user auth and a comparative-analytics dashboard.

Attacking & Defending Confidential VMs

Computer Security, IBA Karachi

- Reproduced a published attack class (HECKLER): an untrusted hypervisor injects interrupts into hardware-isolated Confidential VMs on **AMD SEV-SNP** and **Intel TDX** TEEs, a gap open even when guest memory and CPU state are encrypted.
- Built a testbed on **AMD EPYC** (Milan/Genoa) across three iterations: stock **QEMU**/modified-kernel, a containerized build, then a patched-QEMU instance with a custom **Python** harness over the QEMU Machine Protocol (QMP); characterized guest-kernel behavior under injected NMI and Machine-Check exceptions via forensic **dmesg** analysis.
- Proposed **Guest_Defense**, a guest-side mitigation validating that an interrupt matches the CPU's expected state at minimal overhead (advised by Dr. Faisal Iradat); separately reverse-engineered production Windows software (**Camtasia, VEGAS, 7-Zip**) to map control flow and licensing behavior.

SHA-256 Across Kernel, User Space & Syscalls

Operating Systems, IBA Karachi

- Built **three separate custom Linux kernels** (WSL) to isolate each path, implementing **SHA-256** as a loadable **kernel module** via the kernel crypto API (crypto/hash.h) logging to dmesg, a user-space application over **OpenSSL**, and a **custom system call** exposing kernel cryptographic functionality to user programs.
- Benchmarked all three paths on identical inputs, quantifying the **kernel/user-space boundary-crossing overhead** and isolating where syscall and context-switch costs dominate.
- Analyzed the security trade-offs of each environment: privilege boundaries, access control, and the attack surface of running cryptographic code in **ring 0** versus user space.

Nested Learning & Continual Adaptation

Neural Networks & Deep Learning, IBA Karachi

- Reproduced the core mechanisms of **Nested Learning** (2025), a reformulation of deep networks as a hierarchy of nested optimization processes at different time scales that recasts the optimizer as a memory system and targets **catastrophic forgetting** in deployed models.
- Implemented it from scratch in **PyTorch** (nested fast/medium/slow modules, a **Continuum Memory System** on EMA shadow weights, a **self-modifying meta-learner** that rewrites its own update rule each step), benchmarked against an MLP baseline to confirm optimization behaves as multi-rate memory.

Autonomous Maze-Solving Robot

Arduino & Discrete Logic, 3rd of 45 teams

- Built two independent from-scratch implementations of a maze-solving car on three IR sensors through a dual H-bridge driver: one in embedded C on **Arduino**, one as pure combinational hardware with no microcontroller and no firmware.
- Derived the full truth table and minimized it with **Karnaugh maps** into four Boolean equations in discrete logic ICs; the same equations drive the Arduino firmware unchanged, making the builds identical by construction. Placed **3rd of 45 teams**.

AI Résumé Generator

Introduction to Text Analytics, IBA Karachi

- Fine-tuned **Llama-2 7B** with **LoRA** on 500 résumés, **RAG**-grounded for structure and tone; beat baseline **ChatGPT** head-to-head.

Real-Time E-Commerce Analytics Platform

Big Data Analytics, IBA Karachi

- Built an end-to-end streaming platform over 3-4 GB of high-velocity event data: ingest through **Apache Kafka** into a hybrid **Hadoop/HBase** and **Redis** layer, processing in **Apache Spark**, orchestrated as **Apache Airflow** DAGs and containerized in **Docker** across multiple nodes.
- Shipped three analytics workloads on one live pipeline (**fraud detection**, **customer analytics**, **demand forecasting**) via **Python** dashboards on **Redis**-cached aggregates, plus an operator dashboard exposing live ingestion throughput and HBase query telemetry.

CartNest, Full-Stack E-Commerce Platform

Web Application Development, IBA Karachi

- **React** SPA over **Node.js/Express** and **PostgreSQL** (Docker Compose); **buyer/seller/admin** roles on **JWT** + **bcrypt**, with **Multer** uploads.
- Clean routes/controllers/middleware with SQL migrations and Chart.js analytics.

Student Forum Portal, Full-Stack Community Platform

Database Systems, IBA Karachi

Stack: React, TypeScript, Node.js, SQLite

- Built a full-stack discussion forum (**React/TypeScript** over a **Node.js/TypeScript** + **SQLite** backend): threaded posts, comments, user accounts and content browsing end-to-end.
- Designed a normalized relational schema with enforced **foreign-key constraints** and indexed hot columns; used **parameterized queries** to eliminate SQL-injection risk and **transactions** for multi-step write consistency.

Parallel Matrix Convolution

Parallel & Distributed Computing, IBA Karachi

- Parallelized 2D matrix convolution (the core of image filtering) in **C++** with **OpenMP**, decomposing the input into four quadrants across 8 concurrent threads to cut execution time over the serial baseline.
- Managed per-thread output aggregation and persisted the input, filter and result matrices to separate files, handling the partitioning and recombination that parallel decomposition requires.

NayaPay Wallet Redesign

Human-Computer Interaction, Figma

- **250** high-fidelity **Figma** screens on an 11-component design system across payment, billing and onboarding.
- Full **Urdu** RTL build (124 screens) mirroring the English set, covering 12 end-to-end flows.

SKILLS

Languages: Python, Java, JavaScript / TypeScript, C / C++, Assembly, SQL, Solidity, Verilog, Embedded C (Arduino)

Data Engineering: Apache Spark, Kafka, Airflow, Hadoop, HDFS, HBase, Hive, MongoDB, Redis, PostgreSQL, SQLite

Web & Tools: React, Node.js / Express, MERN / PERN stacks, Django, REST APIs, JWT auth, Tailwind CSS, Docker, Git, Linux, Figma

Security: Bug bounty, web app pentesting, Burp Suite, sqlmap, SQLi Dumper, Nmap, reverse engineering, binary analysis, QEMU, TEEs (SEV-SNP / TDX), CTF authoring, applied cryptography

Blockchain / Web3: Solidity, Ethereum, Foundry, smart-contract auditing & PoC development, contract verification, DApps, blockchain fundamentals

AI / ML: PyTorch, scikit-learn, neural networks, gradient boosting, teacher-student distillation, LoRA fine-tuning, RAG, LLMs (Llama-2, Qwen 2.5, Gemini API), CUDA C/C++

CERTIFICATIONS

- **NVIDIA Deep Learning Institute:** Getting Started with Accelerated Computing in CUDA C/C++ 2025
- **Cisco Networking Academy:** Getting Started with Cisco Packet Tracer; Exploring Networking with Cisco Packet Tracer 2025

ACHIEVEMENTS & LEADERSHIP

- **Winner, Trading Bazaar**, IBA Karachi.
- **IBA Data Science Society:** Designed and authored the cybersecurity-track **CTF** challenges and judged submissions (validating solutions, scoring participants).
- **CryptoHack:** Completed the Introduction, General, Symmetric Ciphers, Diffie-Hellman, and Hash Functions categories.